Area-Efficient and Low-Power 8T Compute-SRAM Bitcell Design for Digital Compute-In-Memory Macros in 22nm CMOS

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***Abstract*—This brief presents a novel compute-SRAM bitcell design capable of performing 1-bit digital multiplication for digital compute-in-memory (DCIM) macros. Unlike the conventional approach, which requires 10 transistors (6T SRAM + 4T NOR), the proposed bitcell achieves the same functionality with only 8 transistors. This is accomplished by efficiently embedding a NOR gate into the latch pair of the traditional 6T bitcell through the addition of two transistors. The design enables efficient in-bitcell compute operations, achieving a 37.5% reduction in area and a 41.5% reduction in leakage compared to the 10T counterpart. Moreover, the proposed bitcell maintains robust stability in terms of write and hold static noise margins, thanks to the sizing strategy and gate overdrive. To validate the design, a 64×64 DCIM macro was fabricated using a 22nm CMOS process. The macro demonstrates a peak energy efficiency of 126 TOPS/W at INT4 precision under a 0.55 V supply.**

***Index Terms*—Compute-in-memory (CIM), digital CIM, SRAM bitcell, high area efficiency, multiply-accumulate (MAC).**

I. Introduction

OVER the last decades, the rapid development of AI has significantly increased the demand for high-performance computing systems capable of handling large-scale data and performing parallel computations, particularly in deep neural networks. In these networks, the majority of computations are concentrated in the convolutional layers, which rely on forward propagation to perform large matrix multiplications, requiring a number of multiply-and-accumulate (MAC) operations[1]. Traditional computing architectures, which separate memory and processing units, face performance bottlenecks, such as high latency and energy consumption, due to frequent data transfers between these units. This issue is commonly referred to as the von Neumann bottleneck. To address these challenges, Compute-in-Memory (CIM)[2] has emerged as a promising solution. CIM integrates MAC operations directly within the memory, using stored values as weights and performing calculations with external input, thereby reducing data movement, latency, and power consumption.

Analog and digital CIM designs are two approaches to addressing the von Neumann bottleneck. In recent years, SRAM has been widely adopted in both ACIM and DCIM macros due to its high speed, scalability, and stability [3], [4], [5], [6], [7], [8], [9], [10], [11], [12], [13], [14], [15], [16], [17]. In SRAM-based ACIM, computations in the voltage, current, or charge domains are typically leveraged to enable parallel multiply-accumulate (MAC) operations. For example, [3] proposes a 9T CIM bitcell that uses a transmission gate alongside a 6T SRAM to implement NOR logic. Although this approach achieves a compact layout, the transmission-gate-driven output suffers from limited drive strength. In [4], [5], [6], [7], inputs are loaded onto word-lines (WLs), while weights are stored in SRAM cells. The MAC operations are executed directly within the SRAM array, producing analog outputs on bit-lines (BLs), which are then converted to digital codes via an ADC. This approach achieves energy- and area-efficient in-memory analog MAC computations, although its accuracy is limited by device variations and analog circuit non-idealities. On the other hand, SRAM-based DCIM relies on in-situ binary logic to perform computations [10], [11], [12], [13], [14], [15], [16], [17]. For instance, [13] proposed a 10T SRAM-based DCIM design that integrates a 4T NOR gate into the 6T bitcell to facilitate 1-bit digital multiplication. The MAC operations are carried out using bit-serial multiplication and parallel adder trees. This approach offers better precision, stability, and scalability compared to ACIM but incurs higher area and power overhead.

To improve area efficiency and reduce leakage power of SRAM-based DCIM, this brief proposes a novel 8T compute-SRAM bitcell. The proposed bitcell integrates 1-bit multiplication capability into the 6T bitcell with only two additional transistors, achieving 37.5% area reduction compared with the conventional 10T approach[13]. In addition, the proposed bitcell architecture allows for minimum transistor sizing in the bitcell latch pairs with similar write/hold stability as conventional 6T bitcell, resulting in 41.5% leakage power reduction. It inherits all the advantages of DCIM while outperforming the conventional 10T SRAM in terms of write noise margin, read power consumption, and read speed. This design was fabricated using 22nm CMOS process, and its functionality and performance metrics were thoroughly evaluated through chip testing.

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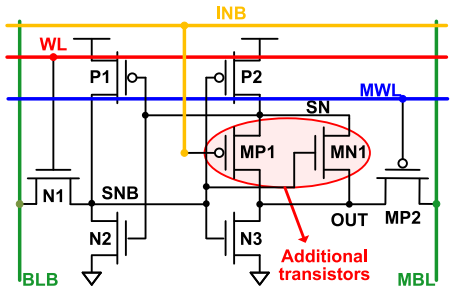


Fig. 1. Schematic of the proposed 8T compute-SRAM.

TABLE I
TRUTH TABLE OF 8T COMPUTE-SRAM OPERATIONS

	BLB	MBL	WL	MWL*	INB*	SN	OUT
WR0	H	L	H	L	L	L	L
WR1	L	H	H	L	L	H	H
KEEP	—	—	L	H	L	—	—
CIM (Read)	—	L	L	IN (H)	INB (L)	—	SN•IN (SN)

*Voltage at L state is VSSL, i.e., -0.3V

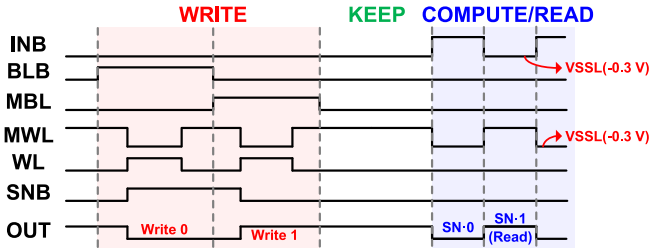


Fig. 2. The timing waveform of the proposed 8T SRAM.

The remainder of this brief is organized into three sections. Section II details the design of the proposed 8T SRAM. Section III describes the CIM macro test chip and presents the measurement results. Section IV concludes this brief.

II. PROPOSED 8T COMPUTE-SRAM BITCELL AND COMPARATIVE ANALYSIS

A. Basic Operation

The schematic of the proposed 8T compute-SRAM is illustrated in Fig. 1. This design builds upon the conventional 6T SRAM by introducing two additional transistors, MP1 and MN1. First, MP1 is incorporated to form an embedded NOR gate (P2, MP1, N3, MP2) within the latch pair. Second, MN1 is added to prevent the node SN from floating when SNB=1 and INB=1. Additionally, one NMOS access transistor is replaced with a PMOS (MP2) to achieve a more compact layout design (see Section II-B). Notably, both MP1 and MP2 require gate overdriving to VSSL (−0.3V) [18] through level shifters to fully turn on. In addition to the conventional WL and BL control signals, new control lines, including MWL (multi-purpose wordline), MBL (multi-purpose bitline), and INB (the inverse of IN), are introduced to enable the WRITE, KEEP, COMPUTE, and READ operations, as detailed in Table I and Fig. 2. Data is stored at node SN, with its complementary value held at node SNB.

During the WRITE operation, the MP2 transistor, controlled by the MWL (effectively acts as the inverse of WL), functions as the access MOS transistor for the bitcell. The MBL serves as the bitline, while INB is set to VSSL to fully activate MP1, ensuring a proper write operation in the latch pair.

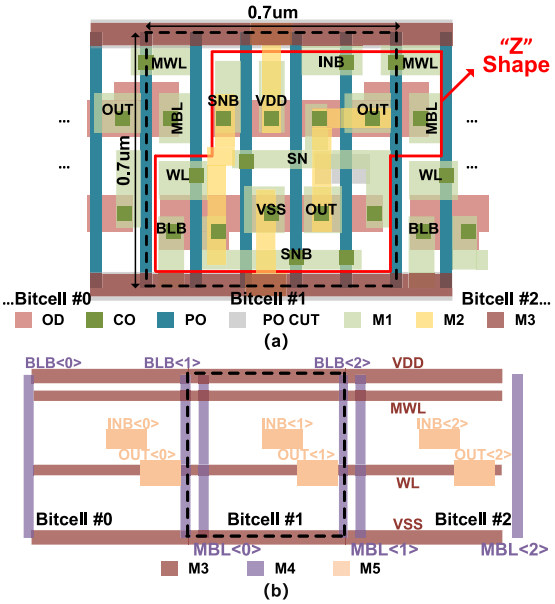


Fig. 3. Layout of the proposed bitcell: (a) "Z"-shape design; (b) three cells with mental layers.

To KEEP the bitcell state, WL is set to VSS, and MWL is tied to VDD to deactivate the access MOS transistors (N1 and MP2). Similar to the write operation, INB is set to VSSL to fully turn on the MP1 transistor.

For COMPUTE operation, both MBL and WL are set to VSS. The MWL is configured to IN, enabling the MP2 transistor to function as the pull-down MOS transistor of a logic NOR gate. This setup ensures the accurate execution of the 1-bit multiplication operation, SN•IN. The result is then aggregated with outputs from other activated words through a parallel adder tree, processed in a column-wise manner.

The READ operation can be viewed as a special case of the compute operation, where only one IN line is activated (equivalent to selecting one word for readout). In this scenario, MBL and WL are set to VSS, while MWL is set to VDD. The INB of the selected bitcell is set to VSSL to perform the multiplication SN•1, which effectively outputs the stored state SN. Since only one word is activated during the read operation, the adder tree can be repurposed as a multiplexer to ensure accurate data retrieval.

B. Sizing and Layout Design

By utilizing the adder tree as a multiplexer for direct digital readout of the selected row, this design not only eliminates the need for dedicated read circuits, saving area, but also avoids the area overhead associated with conventional sizing constraints imposed on access, pull-up, and pull-down transistors for read stability in 6T-SRAM bitcells. Traditionally, to ensure successful write operations and prevent read-disturb, access and pull-down transistors are typically sized 2X and 4X larger than the pull-up transistors, while the pull-up transistors are minimum sized [19], respectively. However, since read operations in this design are performed statically via the adder tree, the pull-down transistors no longer need to provide excess strength to maintain the stored state during reads. As a result, minimum-sized transistors can be used as pull-down transistors

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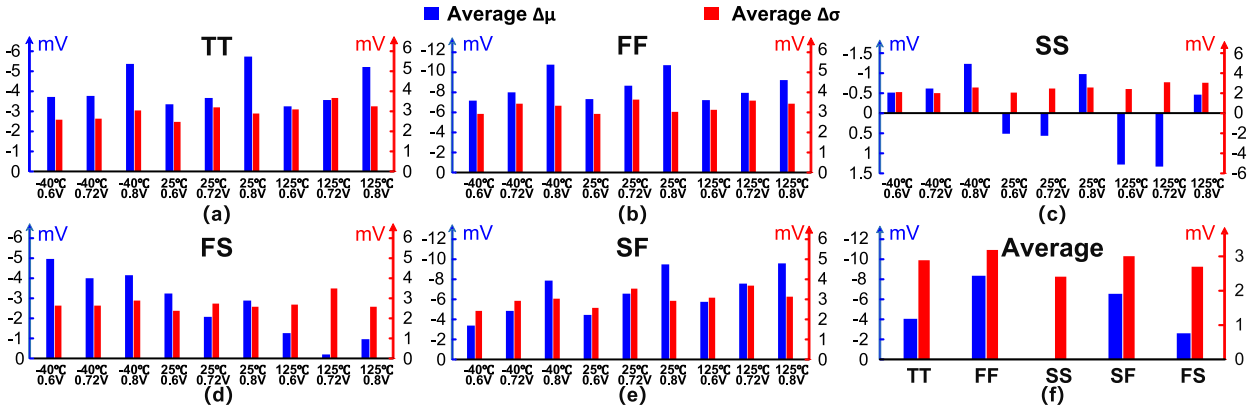


Fig. 4. Monte Carlo simulation results of HSNM showing $\Delta\mu$ ($\Delta\mu$=$\mu_{proposed}$-$\mu_{conv.}$) and $\Delta\sigma$ ($\Delta\sigma$=$\sigma_{proposed}$-$\sigma_{conv.}$) under different temperature and supply voltage: (a) TT; (b) FF; (c) SS; (d) FS; (e) SF; (f) averaged results over all corners.

TABLE II
SUMMARY OF TRANSISTOR SIZE (PRE-SHRINK)

P-type	W(nm)/L(nm)	N-type	W(nm)/L(nm)
P1/P2/MP1	100/30	N2/N3/MN1	100/30
MP2	150/30	N1	150/30

in the bitcell. In this design, only access transistors (N1 and MP2) are upsized to 1.5× the minimum width to ensure stable operation, while all other transistors remain at minimum size, yielding a 12.5% area reduction compared to the traditional SRAM sizing strategy. Table II displays the transistor size details in the proposed bitcell. This approach results in a bitcell area of 0.49 μm^2 in 22 nm technology and low static leakage, representing at least a 37.5% area reduction compared to a conventional 6T SRAM with a 4T NOR gate [13].

The layout design of the proposed 8T SRAM is shown in Fig. 3. The bitcell is arranged in a "Z" shape: NMOS transistors follow the sequence N1→N2→N3→MN1, and PMOS transistors follow P1→P2→MP1→MP2. Uninterrupted active regions maximize area efficiency and enable seamless abutment with neighboring bitcells, yielding a 26.8% area reduction. Furthermore, the "Z" shape layout allows for additional area reduction when the push rules are applied as in commercial SRAM macros.

C. Stability Analysis

The stability of SRAM is typically quantified by three key metrics: RSNM (read static noise margin), HSNM (hold static noise margin), and WNM (write noise margin). However, since the proposed 8T SRAM does not utilize bit line precharging for readout, the RSNM is not applicable for evaluating the stability of this bitcell. Moreover, the 10T bitcell[13] comprises a standard 6T SRAM with a 4T NOR gate connected to the storage node.This added NOR gate contributes negligible capacitance, so both HSNM and WNM remain effectively unchanged as that of conventional 6T bitcell. Therefore, this section focuses on comparing the HSNM and WNM of the proposed 8T bitcell with those of a conventional 6T bitcell to validate its stability.

For HSNM evaluation, Fig. 4 presents extensive Monte Carlo simulations of both the proposed 8T and a conventional 6T bitcell across five process corners (TT, FF, SS, FS, SF), three supply voltage levels (0.6 V, 0.72 V, 0.8 V), and three temperature extremes (–40 °C, 25 °C, 125 °C), thereby delivering a comprehensive, industry-standard PVT-corner reliability assessment. Here, $\Delta\mu$ represents the difference in the average HSNM (including both right- and left-side HSNM) between the 8T and conventional 6T bitcells, while $\Delta\sigma$ indicates the difference in their standard deviations.

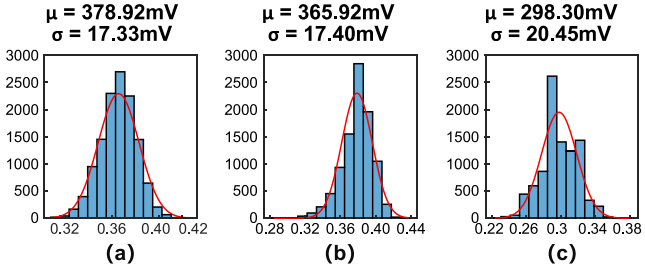
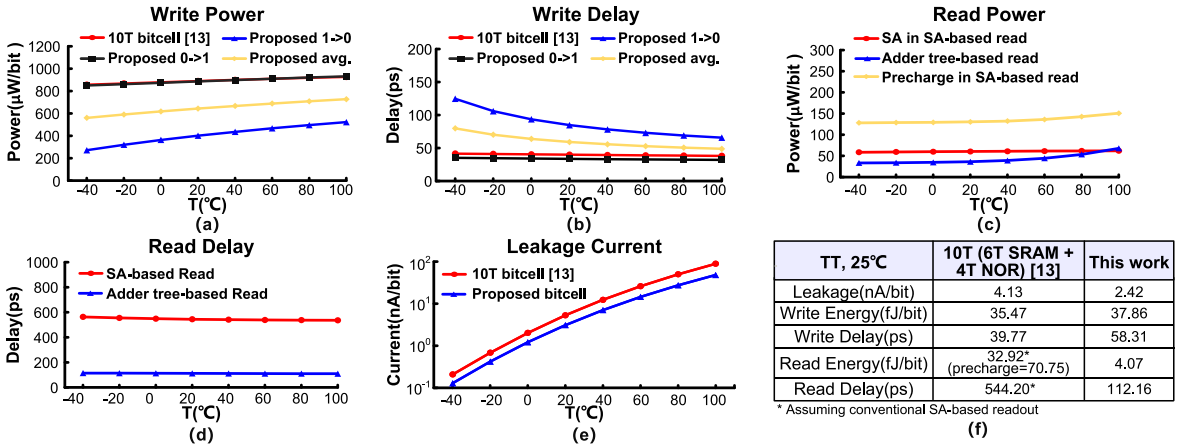


Fig. 5. Monte Carlo simulation results of WNM: (a) writing "0" for the proposed 8T bitcell; (b) writing "1" for the proposed 8T bitcell; (c) conventional 6T bitcell.

The HSNM of the proposed bitcell shows a slight degradation in μ, ranging from −1.3 mV to 10.45 mV, and a worsening in σ, ranging from 2 mV to 3.55 mV, across all corners, voltages, and temperatures. This behavior is attributed to the reduced gain caused by MP1 and MN1 on the right side of the latch pair in Fig. 1, which creates an imbalance in the latch pair's strength. In detail, the worst-case HSNM degradation occurs at the FF corner (Fig. 4(b)), with a maximum reduction of 10.45 mV (0.8 V, 25°C), representing approximately 3.9% of the HSNM in that case. The maximum increase in standard deviation is 3.55 mV (0.72V, 25°C), equivalent to roughly 1.4% of the HSNM. Fig. 4(f) summarizes the average impact across all corners. In the worst-case FF corner, the average $\Delta\mu$ is −8.35 mV (approximately 3% relative to the SNM), and the average $\Delta\mu$ is 3.19 mV (approximately 1.2% relative to the SNM). All other corners exhibit smaller absolute $\Delta\mu$ and $\Delta\sigma$ value. Across 45 conditions (5 corners×3 voltages×3 temperatures), the maximum HSNM degradation remains below 4%, demonstrating nearly identical stability and voltage scalability to the 6T bitcell during static conditions (VSSL=−0.3 V).

Similarly, Fig. 5 presents the Monte Carlo simulation results of WNM (write noise margin) for both the proposed 8T bitcell and the conventional 6T bitcell. Due to the asymmetric design of the proposed bitcell, the WNM for writing "1" and "0" are simulated separately. The results show that the μ of

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TT, 25℃	10T (6T SRAM + 4T NOR) [13]	This work
Leakage(nA/bit)	4.13	2.42
Write Energy(fJ/bit)	35.47	37.86
Write Delay(ps)	39.77	58.31
Read Energy(fJ/bit)	32.92* (precharge=70.75)	4.07
Read Delay(ps)	544.20*	112.16

* Assuming conventional SA-based readout

(f)

Fig. 6. Bitcell-level simulation results: (a) write power; (b) write delay; (c) read power; (d) read delay; (e) static leakage; (f) summary at TT and 25°C.

WNM for writing "0" and "1" in the proposed 8T bitcell are 378.92 mV and 365.92 mV, respectively, representing improvements of 27% and 23% compared to the conventional 6T bitcell. Additionally, the proposed 8T bitcell achieves a 15% reduction in σ of WNM. All process corners exhibit at least 20% WNM improvements. These improvements in WNM are attributed to the sizing strategy applied to the latch pairs and access transistors, which facilitates easier flipping of the latch pairs during write operations.

D. Delay and Power Analysis

Fig. 6 presents the bitcell-level simulation results of static leakage current, write/read power consumption, and delay at 0.8 V across various temperatures for both the proposed 8T bitcell and the 10T bitcell [13] (effectively conventional 6T bitcell [19] + 4T NOR). Due to the asymmetry of the proposed 8T bitcell, write power consumption and delay were simulated for both transitions: flipping from 1 to 0 and from 0 to 1. Fig. 6(a) shows that the average write power consumption of the proposed 8T bitcell is 21.5%-34.6% lower than that of the conventional bitcell, attributed to the reduced capacitance of the minimum-sized latch pairs. For write delay in Fig. 6(b), the proposed bitcell exhibits varying performance. When flipping from 1 to 0, the weak pull-up of access NMOS (N1) and weak pull-down of access PMOS (MP2) result in lower current and longer access time. Conversely, when flipping from 0 to 1, the strong driving strength and larger size of the access transistors reduce write delay compared to the conventional bitcell. Overall, the proposed bitcell has a slightly higher average write delay than the conventional 6T bitcell.

Fig. 6(c) and (d) show the read power consumption and delay, respectively. The adder tree-based read approach used in this work achieves up to 43.1% lower read power consumption and 79.5% lower read delay compared to the conventional sense amplifier (SA)-based [20] readout. When including the power consumption of BL precharge, the proposed 8T bitcell exhibits up to 82.1% lower read power consumption. The adder tree-based readout scheme eliminates bitline precharging: when reading a '1', only few least significant bits of the adder tree toggle; when reading a '0', only static energy is consumed. In contrast, the SA-based read must precharge the bitline before sensing, resulting in higher energy due to the large parasitic capacitance of bitlines. Fig. 6(e) shows up to 46.1% lower leakage current in the proposed bitcell compared to the conventional bitcell, thanks to the use of minimum-sized transistors and stacking transistors in the latch pair.

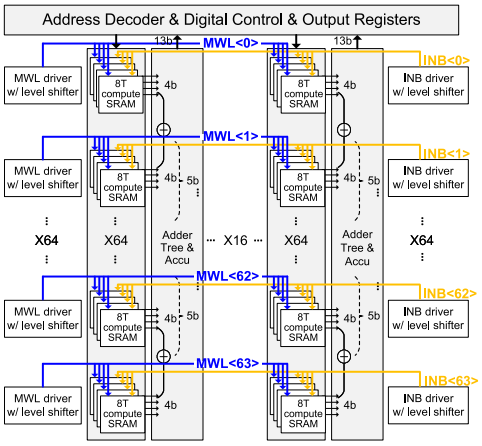


Fig. 7. Macro-level testchip design.

To summarize, Fig. 6(f) compares the key metrics at TT corner and 25°C. Compared with the conventional 6T bitcell, the proposed bitcell achieves a 41.5% reduction in static leakage current, an 88.6% reduction in read energy, and a 79.4% reduction in read delay, while maintaining similar write energy and incurring a 32.0% increase in write delay.

III. Testchip Design and Measurements

To validate the proposed bitcell, a DCIM macro based on the proposed bitcell was designed and fabricated using 22nm CMOS technology. Fig. 7 illustrates the macro structure, which consists of a 64×64 array of bitcells. Each 64×4 subarray is alternately arranged with an adder tree and accumulator, following a configuration similar to [13]. The MWL and INB drivers, equipped with embedded differential cascode voltage switch (DCVS)-based level shifters (which incurs nearly 3.11% dynamic power consumption), are positioned on the left and right sides of the array, respectively. The address decoder and digital control module generate the necessary control signals based on the SRAM operation mode. During compute mode, each subarray performs the INT4 MAC operations in 5 clock cycles as in [13], and the 13-bit results are then forwarded to the output register.

Thanks to the proposed 8T bitcell, which achieves a 37.5% reduction in area and 41.5% reduction in leakage power at the

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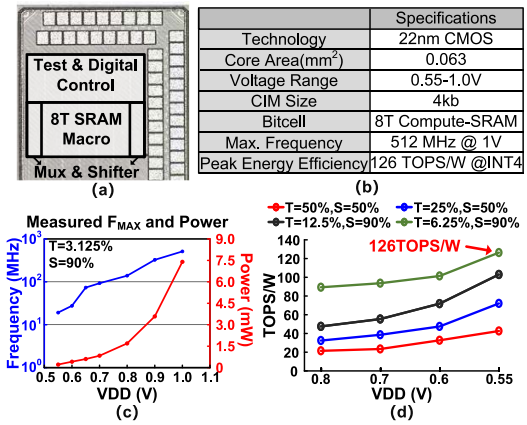


Fig. 8. Testchip summary: (a) chip micrograph; (b) specifications; (c) measured maximum frequency and power vs. supply voltage VDD; (d) measured energy efficiency vs. supply voltage, toggle rate, and sparsity.

TABLE III
COMPARISON WITH PRIOR SRAM-BASED DCIM MACROS

	ISSCC'21 [13]	ISSCC'22 [15]	ISSCC'23 [16]	ISSCC'25 [17]	This work
Technology	22nm	5nm	4 nm	28 nm	22nm
MAC operation	Digital	Digital	Digital	Digital	Digital
Bitcell type	6T SRAM + 4T NOR	12T SRAM + shared NOR	8T x 2bit + OAI	6T + shared 4T AND	8T compute-SRAM
Array size	64Kb	64Kb	54 Kb	32 Kb	4Kb
Bitcell area (um²)	0.784(1)	0.075	0.161 (2bit)	1.094(1)	0.49
Normalized bitcell area (F²)(2)	1620	3000	5031	1396	1012
Push rule in bitcell?	No	No	No	No	No
Full MAC precision?	Yes	Yes	Yes	Yes	Yes
VDD (V)	0.52-0.92	0.5-0.9	0.32-1.1	0.55-0.9	0.55-1.0
Energy efficiency (TOPS/W)	89 (4b/4b)	254 (4b/4b)	99.6-349.6(3) (4b/4b)	70.2-200 (4b/4b)	126 (4b/4b)

(1) Estimated from the layout shown in the figure of the article
(2) Calculated by Area/(Feature Size)² (3) Normalized to 4b/4b

cell level, the overall macro-level area and leakage power in our test chip are reduced by 7.5% and 9.30%, respectively, given that the bitcell array accounts for 20.01% of the total area and 22.40% of the total leakage power. These benefits are further enlarged when approximate adder tree architectures with reduced area are employed.

Fig. 8 summarizes the testchip specifications and measurement results. The macro occupies an area of 0.063 mm^2 and achieves a maximum operating frequency of 512 MHz at a 1.0 V supply, consuming 7.4 mW of power under 90% data sparsity and 3.125% toggle rate. The peak energy efficiency of 126 TOPS/W is attained at 0.55 V, with an input toggle rate of 6.25% and a weight sparsity of 90%. Table III compares the proposed design with state-of-the-art SRAM-based CIM macros. Our proposed design achieves the lowest normalized area of 1012 F^2, representing a 27.5%-79.9% reduction compared to prior arts.

IV. CONCLUSION

In this brief, a 8T compute-SRAM that integrates the functionality of a 6T SRAM and a 4T NOR gate has been introduced. Compared to prior DCIM [13], the proposed bitcell achieves a 37.5% reduction in area, 41.5% reduction in static power consumption, 88% reduction in read power. Thanks to the sizing strategy and gate overdrive, WNM is improved by 22.7% with less than 4% decrease in HSNM. Based on this bitcell, a 64×64 macro fabricated in 22nm CMOS demonstrates a peak energy efficiency of 126 TOPS/W.

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